

1. How to clone a GitHub repository

`git + clone + URL`

How to use the git commands

`git + pull:`

`git + add + .`

`git + commit + -m + label for your changes here`

`git + push:`

How to write a Markdown file that contains images and proper formatting

You save the file as .md file. This is how you insert images src.

How to convert a Markdown file to PDF

You right-click and click the Markdown PDF: export (pdf) option

2. How to compress (zip) a directory/folder in Debian

Go to files, right click the zip file and press extract.

3. What are Absolute paths and relative paths?

Absolute path - The full address of the file. Relative paths - identifies the location of the directory.

4. How to work with the manual pages (man command)

`man + option`

5. How to parse (search) for specific words in the manual page

`grep "word" file`

6. How to redirect output (>, >>, and |)

7. How to append the output of a command to a file

8. How and when to redirect the output of a command to another (pipes)

`cat file.txt | grep "error"`

9. How to use echo and output redirection to create a new file that contains some text

`echo "hi" > file`

10. How to use wildcards (For copying and moving multiple files at the same time)

You can use the . `mv ~/Downloads/Nature/ ~/Pictures/wallpapers/ cp ~/Downloads/homework/.pdf ~/Downloads/.txt ~/Projects/school/`

11. How to use brace expansion (For creating entire directory structures in a single command)

`mkdir -pv project_jupiter/site/{old,new}/{code/{scripts,markup},assets/{img,mp3,mp4}}`

12. How to create a simple “hello world” shell script

Echo "hello world"

13. How to use variables in a shell script

Assign it using a = operator and to refer to it put a \$ before the variable name

14. For each of the following commands, include a definition, syntax/formula/usage/, and 2 - 5 well-documented examples

* AWK

- **Definition:**
 - Awk is used for displaying and processing text.
- **Formula:**
 - `awk + options + {awk command} + file + file to save (optional)`
- **Examples:**
 - Print first field of /etc/passwd file.
 - `awk + -F: + '{print $1}' + /etc/passwd`
 - Print the last field of the /etc/passwd file.
 - `awk + -F: + '{print $1}' + /etc/passwd`
 - Print the first and last field of the /etc/passwd.
 - `awk + -F: + '{print $1, " = ", $NF}' + /etc/passwd`
 - Prints the length of a line(record),
 - `awk + '{print length($0)}' + /etc/passwd`
 - Start printing a file from a given line.
 - `awk + 'NR > 3 + { print }' + /etc/passwd`

* CAT

- **Definition:**
 - The cat command is used for displaying the content of a file.
- **Formula:**
 - `cat + option + file(s) to display`
- **Examples:**
 - Display the content of a file located in ~/Documents/sample_files/Code/helloWorld.py
 - `cat + ~/Documents/sample_files/Code/helloWorld.py`

- Display the content of a file.
 - `cat + -n + ~/Documents/sample_files/Code/helloWorld.py`

* CP

- **Definition**
 - Copies files and directories from a source to a destination.
- **Formula:**
 - `cp + files to copy + destination`
- **Examples:**
 - To copy directories you must use the -r option.
 - `cp + -r + directory to copy + destination`
 - To copy a directory with absolute path.
 - `cp + -r + ~/Downloads/wallpapers + ~/Pictures/`
 - To copy the content of a directory to another directory.
 - `cp + Downloads/wallpapers/* + ~/Pictures/`
 - To copy a file.
 - `cp + Downloads/wallpapers.zip + Pictures/`
 - To copy multiple files in a single command.
 - `sudo + cp + -r + script.sh + program.py + home.html + assets/ + /var/www/html/`

* cut

- **Definition**
 - Used to extract a specific section of each line of a file and display it to the screen.
- **Formula:**
 - `cut + option + files`
- **Examples:**
 - Display a list of all the users in your system.
 - `cut + -d + ':' + -f1 + /etc/passwd`
 - Display a list of all the users in your system with their login shell.
 - `cut + -d + ':' + -f1,7 + /etc`

* grep

- **Definition**
 - Grep is used to search text in a given file.
- **Formula:**
 - `grep + option + search criteria + file(s)`
- **Examples:**
 - Search any line that contains the word "dracula" in the given file.
 - `grep + 'dracula' + ~/Documents/dracula.txt`
 - Search any line that contains the word 'dracula' regardless of the case.
 - `grep + -i + 'dracula' + ~/Documents/Books/dracula.txt`
 - Display how many lines contain the matched string.
 - `grep + -c + 'dracula' + ~/Documents/Books/dracula.txt`
 - Search for all the lines that do not contain the word 'war'.

- `grep + -v + 'war' + ~/Documents/Books/war-and-peace.txt`
- Display your user's information as stored in the `/etc/passwd`.
 - `grep + -i + $USER + /etc/passwd`

* head

- **Definition**
 - The head command displays the top N number of lines of a given file.
- **Formula:**
 - `head + option + file(s)`
- **Examples:**
 - Display the first 10 lines of a file.
 - `head + ~/Documents/Book/dracula.txt`
 - Display the first 5 lines of a file.
 - `head + 5 + ~/Documents/Book/dracula.txt`
 - Display the first 5 lines of multiple files.
 - `head + -n + 5 + Txt/{dracula,war-and-peace}.txt`
 - Display the first line of multiple files using wildcards.
 - `head + -n + 1 + Csv/*.csv Code/*.py`
 - Display a given number of lines of the output of a given command.
 - `ls + -l + ~/cis106/ + | + head + -n + 2`

* ls

- **Definition**
 - List files.
- **Formula:**
 - `ls + directory`
- **Examples:**
 - Long list including hidden files.
 - `ls + -al + directory/to/list`
 - List all files classified.
 - `ls + -R + directory/to/list`

* man

- **Definition**
 - An interface to the system reference manuals.
- **Formula:**
 - `man + option + command`
- **Examples:**
 - How to display debugging.
 - `man + -d`
 - How to display the source nroff file.
 - `man + -w`

* mkdir

- **Definition**

- Used for creating directories.

- **Formula:**

- `mkdir + the name of the directory`

- **Examples:**

- Create a directory in the present working directory
 - `mkdir + wallpapers`
- Create a directory in a different directory using relative path.
 - `mkdir + wallpapers/forest`
- Create a directory in a different directory using absolute path.
 - `mkdir + ~/wallpapers/forest`
- Create a directory with a single quote in the name.
 - `mkdir + wallpapers/"majora's mask"`
- Create multiple directories
 - `mkdir + wallpapers/cars + wallpapers/cities + wallpapers/forest`

* rm

- **Definition**

- Removes files and directories.

- **Formula:**

- `rm + option`

- **Examples:**

- Remove a file.
 - `rm + list`
- Remove an empty directory.
 - `rmdir + Downloads/games`
- Remove an non-empty directory.
 - `rm + -r + Downloads/games`
- Remove a file and prompt confirmation before removal.
 - `rm + -i + list`
- Remove all the files inside a directory and ask before removing more than 3 files.
 - `rm + -I + Downloads/games/*`

* mv

- **Definition**

- Moves and renames directories

- **Formula:**

- `mv + source + destination`

- **Examples:**

- To rename a file.
 - `mv + homework.docx + cis106homework.docx`
- To move a file from a directory to another using relative path.
 - `mv + Downloads/Homework.pdf + Documents/`

* TAC

- **Definition**
 - Displays the contents of a file.
- **Formula:**
 - `tac + option + file(s) to display`
- **Examples:**
 - Display the content of a file located in ~/Documents/sample_files/ in reverse order.
 - `tac + ~/Documents/sample_files/Code/helloWorld.py`
 - Display the content of multiple files in reverse order.
 - `tac + ~/Documents/sample_files/Code/helloWorld.py +
~/Documents/sample_files/Code/helloWorld.py`

* Tail

- **Definition**
 - Shows the last N number of lines of a file.
- **Formula:**
 - `tail + option + file(s)`
- **Examples:**
 - Display the last 10 lines of a file.
 - `tail + ~/Documents/sample_files/`
 - Display the last 5 lines of a file.
 - `tail + -5 + ~/Documents/sample_files/`
 - Display the first 5 lines of multiple files.
 - `tail + -n + 5 + Txt/{dracula,war-and-peace}.txt`
 - Display the first line of multiple files using wildcards.
 - `tail + -n + 1 + Csv/*.csv Code/*.py`
 - Display a given number of lines of the output of a given command.
 - `ls + -l + ~/cis106/ + | + tail + -n + 2`

* touch

- **Definition**
 - Used to create files.
- **Formula:**
 - `touch + option + argument`
- **Examples:**
 - To create a file called list.
 - `touch + list`
 - To create a file using absolute path.
 - `touch + ~/Downloads/games.txt`

* tr

- **Definition**
 - Translates standard input to output
- **Formula:**
 - `Standard + output + | tr + option + set + set`
- **Examples:**

- Translate on character to another.
 - `cat file.txt + | + tr + '.' + ',' +`
- Translate tabs into space.
 - `cat file.txt + | + tr + -s + "[:space:]" + ' '`

* tree

- **Definition**
 - list contents of directories
- **Formula:**
 - `Tree + option`
- **Examples:**
 - List contents of directory
 - `Tree`
 - List contents of lab9.
 - `Tree + lab9`